

Benchmarking Large Language Models for Natural Language Processing Tasks:

*A Comprehensive Evaluation of Summarization, Question Answering, and Sarcasm Detection

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Abstract—Large Language Models (LLMs) continue to advance natural language processing, but their performance remains uneven across languages. Arabic, spoken natively by over 400 million people, poses distinct challenges due to complex morphology, orthographic ambiguity, and dialectal variation. This paper presents an open-source benchmarking framework to evaluate multilingual LLMs across three Arabic tasks: text summarization, question answering, and sarcasm detection. We evaluate nine models—Llama-3.3-70B, Kimi-K2, GPT-OSS-20B, GPT-OSS-120B, Qwen3-32B, GLM-5, Gemma-3-27B-IT, DeepSeek-V3.2, and Mistral-Saba—across more than 22,000 dataset instances using lexical metrics (ROUGE, BLEU, METEOR) and neural semantic metrics (BERTScore, Gemini-2.5-Flash LLM-as-Judge). The results show clear, task-dependent differences: Llama-3.3-70B achieves the highest lexical overlap in summarization (ROUGE-1 = 0.3587), GLM-5 leads question answering (ROUGE-1 = 0.6095), and GLM-5 delivers the most balanced performance on dialectal sarcasm detection, with GPT-OSS-20B close behind. No single model leads across all tasks, demonstrating that general leaderboards cannot replace task-specific testing. We provide a parameter-configurable React.js interface to help practitioners evaluate and select models for Arabic deployment.

I. INTRODUCTION

Large Language Models have changed natural language processing [1], [2]. By scaling parameters and training on large text corpora, modern foundation models show strong cross-domain reasoning, translation, and text generation abilities [3]–[6]. However, model performance remains uneven across languages [5]. Arabic has over 400 million native speakers, yet it remains underrepresented in foundation model research [7], [8].

This performance gap stems from both linguistic properties and data scarcity [9]. Arabic morphology uses a non-concatenative root-and-pattern system where a single root yields dozens of surface forms through affixation and vowel patterns [10]. In addition, Arabic is written right-to-left, often omits short vowels, and exhibits sharp differences between formal Modern Standard Arabic (MSA) and regional dialects such as Egyptian, Levantine, and Gulf Arabic [11], [12]. Because public pretraining data is mostly English, models receive limited exposure to Arabic syntax, as well as its colloquial usage [5], [13].

Prior evaluation efforts provide valuable findings but have notable limitations [7], [8]. Many studies evaluate only two or three proprietary models, limiting architectural comparisons [11]. Most rely on one-off scripts that cannot easily be repurposed. Furthermore, surface metrics such as ROUGE and BLEU often penalize valid Arabic paraphrases because they only track exact word matches [14]. Finally, existing tools usually require custom code, making it difficult for practitioners to evaluate candidate models on their own domain data.

To address these challenges, this study presents a standardized, multi-model evaluation framework for Arabic NLP. Our primary contributions are:

- 1) **Broad Empirical Evaluation:** We evaluate nine open-weights and API-accessible models across three distinct Arabic NLP tasks using a standardized protocol with over 22,000 test cases [15].
- 2) **Dual-Metric Diagnostics:** Alongside lexical metrics (ROUGE, BLEU, METEOR), we incorporate semantic metrics (BERTScore and Gemini-2.5-Flash LLM-as-Judge) to assess meaning independently of surface inflection [16]–[20].
- 3) **Configurable Benchmarking Tool:** We release a modular React.js interface that lets researchers and practitioners configure tasks, models, decoding settings, and prompts, generating exportable reports without writing code.

II. BACKGROUND AND RELATED WORK

a) *Multilingual LLMs and Arabic:* Transformer-based models have achieved strong general performance across many natural language tasks [1], [2], [4], [21]. However, their performance often drops on morphologically rich and lower-resource languages because training corpora heavily favor English [7], [9]. In Arabic, non-concatenative morphology and dialectal divergence create severe evaluation hurdles [10], [11]. Everyday communication uses regional dialects that differ substantially from formal Modern Standard Arabic in phonology, vocabulary, and syntax [12], [22]. To address these challenges, several Arabic-centric models have been introduced: Jais was pretrained from scratch on Arabic and English [15], AceGPT

adapted an open backbone using localized instruction tuning [23], and ALLaM paired vocabulary expansion with staged bilingual pretraining [24]. These models highlight the value of language-specific adaptation, providing a comparison point for general multilingual architectures.

b) Arabic Benchmarking Efforts: Standardized benchmarks are critical for tracking progress in Arabic NLP [7]. LAraBench evaluated models across 33 tasks, focusing primarily on classification rather than long-form generation [8]. AlGhafa introduced question answering and generation tasks, but omitted dialectal irony and provided no interactive interface [9]. GPTaraEval analyzed ChatGPT across diverse Arabic tasks, but relied on proprietary APIs whose underlying weights change over time [11]. More recently, ArabicMMLU introduced an exam-based benchmark for Modern Standard Arabic [25], AraDiCE evaluated dialectal and cultural understanding across Egyptian, Syrian, and Gulf Arabic [26], and 3LM benchmarked reasoning in STEM and code generation [27]. Our framework complements these efforts by combining generative and classification tasks under a uniform lexical and semantic evaluation protocol with open, interactive tooling.

c) Evaluation Metrics for Arabic: Evaluating machine-generated Arabic requires combining lexical, semantic, and classification metrics [7], [14]. Surface metrics like ROUGE [16], BLEU [17], and METEOR [18] measure word and phrase overlap. However, because Arabic words inflect heavily, valid paraphrases often share few surface tokens with reference texts [10]. Continuous semantic metrics solve this: BERTScore uses contextual embeddings to match token meanings [19], while LLM-as-Judge prompts an independent model to score overall semantic equivalence [20]. For classification tasks with class imbalance, such as sarcasm detection, precision, recall, macro-F1, and ROC-AUC are necessary to evaluate performance accurately [22], [28].

III. BENCHMARKING FRAMEWORK

A. System Architecture

The framework uses four decoupled layers to separate concerns and support new tasks and models:

1) *Data Layer:* Handles dataset ingestion, schema validation, and Arabic text normalization. Text normalization standardizes Unicode variations (such as distinct representations of alef, taa marbuta, and yaa), strips extraneous control characters, and prepares balanced evaluation samples.

2) *Model Interface Layer:* Connects to inference endpoints (OpenRouter and Groq) through a unified API gateway. This layer handles authentication, payload formatting, rate-limiting backoffs, and retries, allowing new architectures to be added without modifying core evaluation logic.

3) *Task Evaluation Layer:* Executes task prompts, cleans responses, and computes lexical, embedding, and LLM-as-Judge scores.

4) *Reporting Layer:* Exports evaluation metrics to structured JSON summaries, per-sample Excel workbooks, and an interactive dashboard.

The end-to-end pipeline is shown in Fig. 1.

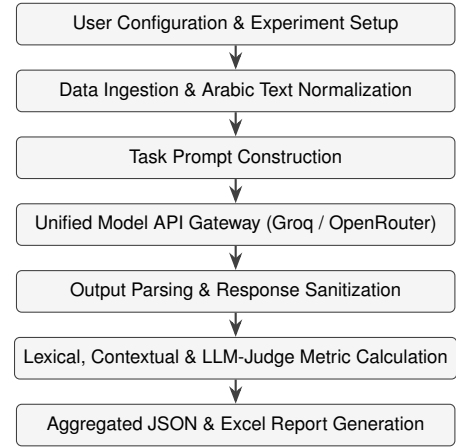


Fig. 1. End-to-end pipeline and architectural layers of the Arabic LLM benchmarking framework.

B. Task Specifications and Prompt Protocols

The framework evaluates three representative NLP tasks:

Task 1: Arabic Text Summarization. Models receive a Modern Standard Arabic news article and are prompted to write an informative summary of at most three sentences. Evaluation combines lexical overlap (ROUGE-1, ROUGE-L, BLEU, METEOR) with semantic matching (BERTScore-F1, Gemini-2.5-Flash LLM-as-Judge).

Task 2: Arabic Question Answering. Models receive a context passage and question, with instructions to provide a concise factual answer grounded strictly within the context. The evaluation uses the same dual-metric lexical and semantic suite.

Task 3: Arabic Sarcasm Detection. Models receive a social media post written in Modern Standard Arabic or Egyptian dialect and classify it as *sarcastic* or *not sarcastic* without extra text. Evaluation uses accuracy, macro-F1, precision, recall, and ROC-AUC.

C. Decoding Hyperparameters

Text generation parameters are standardized across all models:

- *Temperature (T):* Controls the randomness of the model's token selection during generation. Lower values make the model more deterministic by favoring higher-probability tokens, while higher values increase diversity and allow less probable tokens to be selected. The temperature was set to $T = 0.2$ for summarization and question answering to permit natural phrasing while maintaining semantic consistency. For sarcasm detection, $T = 0.0$ (greedy decoding) was used to make the binary classification output as deterministic and reproducible as possible.
- *Maximum Output Length (max_tokens):* Defines the maximum number of tokens the model is allowed to generate in a single response. A token may represent a complete word, part of a word, punctuation, or another text unit depending on the tokenizer. The value was fixed at 2,048 tokens

for all tasks, providing sufficient capacity for complete summaries and question-answering responses while placing an upper bound on generation length and reducing the risk of excessively long or repetitive outputs.

IV. EXPERIMENTAL SETUP

A. Benchmark Datasets

We evaluate models on three public Arabic NLP datasets:

1) *Arabic Text Summarization* [29]: Contains 8,380 article-summary pairs from formal Arabic news publications. Each entry includes a full news report and a human-written abstractive reference summary in Modern Standard Arabic.

2) *ArabicaQA* [30]: Contains over 5,000 factual question-passage-answer triplets from Arabic Wikipedia, spanning history, science, geography, and general knowledge.

3) *Ar_Sarcasm* [28]: Contains 8,440 manually labeled social media posts in Modern Standard Arabic and Egyptian Arabic, testing colloquial and pragmatic understanding.

B. Evaluated Models

Nine open-weights and API-accessible multilingual models were selected to cover diverse parameter scales and architectures, as summarized in Table I.

TABLE I

OVERVIEW OF EVALUATED LARGE LANGUAGE MODELS, INCLUDING DEVELOPER ORGANIZATION, PARAMETER SCALE/ARCHITECTURE, API INFERENCE ACCESS POINT, AND PRIMARY TECHNICAL SOURCE.

Model ID	Organization	Architecture / Size	Access Gateway	Source
Llama-3.3-70B-Versatile	Meta	Dense (70B)	Groq	[31]
Kimi-K2-Instruct	Moonshot AI	MoE (~1T total)	OpenRouter	[32]
GPT-OSS-20B	OpenAI	Dense (20B)	OpenRouter	[33]
GPT-OSS-120B	OpenAI	Dense (120B)	OpenRouter	[33]
Qwen3-32B	Alibaba	Dense (32B)	OpenRouter	[34]
GLM-5	Zhipu AI	MoE (parameter count not publicly disclosed)	OpenRouter	[35]
Gemma-3-27B-IT	Google	Dense (27B)	OpenRouter	[36]
DeepSeek-V3.2	DeepSeek	MoE (~671B total)	OpenRouter	[5]
Mistral-Saba	Mistral AI	Dense (24B)	OpenRouter	[37]

The models pair dense networks (20B to 120B parameters) with large-scale Mixture-of-Experts (MoE) architectures [3], [5]. All nine models include Arabic within their multilingual training data. Citations in Table I point to the technical report or official model card corresponding to each evaluated model checkpoint.

C. Evaluation Metrics

Evaluation is organized across four diagnostic tiers:

1) *Lexical Overlap*: These metrics evaluate how much the generated text resembles the reference (human-written) text in terms of words and word sequences. They are useful for measuring whether the model preserves important content and phrasing from the reference answer.

- *ROUGE-1*: Measures the proportion of individual words (unigrams) from the reference text that also appear in the generated text. In simple terms, it answers: “How many of the important words in the reference did the model include?” A higher ROUGE-1 score indicates greater word-level content overlap with the reference. It is particularly useful for checking whether the generated summary retains the main information from the original text [16].

- *ROUGE-L*: Measures similarity based on the Longest Common Subsequence (LCS), which identifies the longest sequence of words that appears in both the generated and reference texts while preserving their order. Unlike ROUGE-1, which looks at individual words independently, ROUGE-L considers how words are arranged relative to one another. A higher score therefore indicates that the generated text preserves more of the reference’s wording and overall sentence structure [16].
- *BLEU*: Measures how many word sequences (n -grams) in the generated text also occur in the reference text. It therefore focuses more on precision than recall: in simple terms, it asks, “How much of what the model generated is also found in the reference?” BLEU also applies a brevity penalty to prevent very short outputs from receiving artificially high scores. Higher BLEU scores indicate greater similarity in the wording and phrasing of the generated text to the reference [17].
- *METEOR*: Measures the similarity between the generated and reference texts by matching words while also considering stemming and synonyms. This makes it less dependent on exact word matching than metrics such as ROUGE-1 and BLEU. In Arabic, where the same meaning can be expressed using different morphological forms, stemming and synonym matching can help recognize semantically related words even when their surface forms differ. A higher METEOR score indicates greater lexical and meaning-level correspondence with the reference [18].

2) *Contextual Embedding Alignment*: BERTScore measures the semantic similarity between the generated and reference texts by comparing their tokens using contextual embeddings from `bert-base-multilingual-cased`. Unlike exact word-matching metrics, it can give credit when the model expresses the same meaning using different words or phrasing, making it useful for evaluating valid paraphrases [19].

3) *LLM-as-Judge Protocol*: Gemini-2.5-Flash acts as an independent judge model [38]. Since the judge does not share weights with any of the nine evaluated models, the risk of self-enhancement bias is reduced [20]. Candidate and reference answers are evaluated using a standardized rubric, producing a semantic similarity score from 0.0 (complete contradiction) to 1.0 (complete equivalence). This provides an additional assessment of whether the generated answer conveys the intended meaning of the reference, even when the wording differs substantially.

4) *Classification Metrics*: Sarcasm detection is evaluated using standard classification metrics: Accuracy measures the overall proportion of correctly classified examples; Precision measures how often predicted sarcastic examples are actually sarcastic; Recall measures how many of the truly sarcastic examples are correctly identified; Macro-F1 summarizes the balance between Precision and Recall across both classes by giving each class equal weight; and ROC-AUC measures the model’s ability to distinguish between sarcastic and non-sarcastic examples across different classification thresholds. Macro-F1 and ROC-AUC are emphasized as primary metrics

because they provide a more informative assessment than Accuracy when class distributions are imbalanced [28].

V. RESULTS AND CROSS-MODEL ANALYSIS

Evaluating nine language models across the combined test collection reveals clear task-dependent performance variations. Model performance differs across tasks, input domains, and metric families.

A. Task 1: Arabic Text Summarization

Llama-3.3-70B achieves the highest scores across the lexical and embedding-based metrics, leading in ROUGE-1 (0.3587), ROUGE-L (0.2912), BLEU (0.1258), and BERTScore-F1 (0.6680). GLM-5 ranks second on the lexical overlap metrics and achieves the highest METEOR score (0.3080), suggesting strong correspondence with the reference wording, including matches that may benefit from METEOR’s stemming and synonym handling.

However, the LLM-as-Judge evaluation reveals an informative divergence between surface-level overlap and semantic preservation. GPT-OSS-120B achieves the highest Semantic Match score (0.913), despite ranking fourth on ROUGE-1 (0.3400). This suggests that its summaries may preserve the reference meaning while using alternative vocabulary or phrasing that receives lower scores from n -gram-based metrics. In contrast, Mistral-Saba achieves a moderate ROUGE-L score (0.2644) but records the lowest Semantic Match score (0.599), illustrating that relatively strong lexical similarity does not necessarily correspond to strong semantic alignment with the reference. Llama-3.3-70B likewise achieves high lexical scores but a comparatively modest Semantic Match score (0.618), further demonstrating that lexical overlap alone does not guarantee semantic fidelity.

B. Task 2: Arabic Question Answering

Question answering produces a different ranking than summarization. GLM-5 and Gemma-3-27B-IT share top performance on ROUGE-1 (0.6095 and 0.6094), with GLM-5 leading in METEOR (0.5502) and Gemma-3-27B-IT leading in BLEU (0.2403). DeepSeek-V3.2 achieves the highest Semantic Match score (0.861) and BERTScore-F1 (0.7864), showing accurate factual alignment with reference answers.

Across all models, QA scores are substantially higher than summarization scores, with ROUGE-1 ranging from 0.5511 to 0.6095. This increase occurs because question answering requires extracting specific factual spans from context passages, resulting in greater token overlap with references. Notably, Llama-3.3-70B drops to fourth place, and both GPT-OSS models fall to the lower half of the ranking, showing that abstractive summarization and extractive question answering rely on distinct model strengths.

C. Task 3: Arabic Sarcasm Detection

Sarcasm detection shows the widest performance spread, with accuracy ranging from 0.340 to 0.829. GLM-5 achieves the highest Accuracy (0.829) and F1-score (0.538), indicating

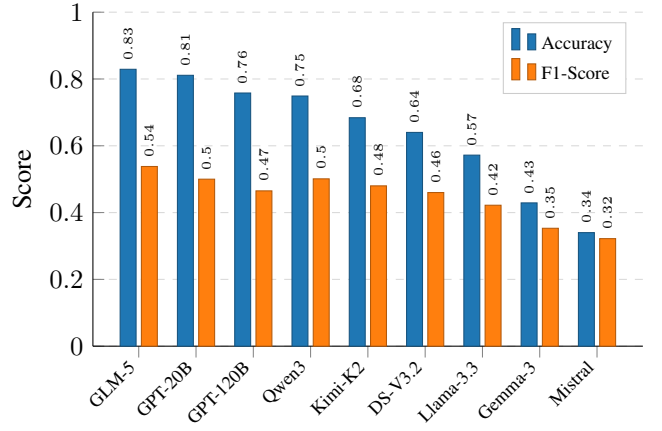


Fig. 2. Sarcasm detection comparison (Accuracy vs. F1-score) across the nine evaluated models, illustrating precision-recall imbalance.

the most balanced performance between correctly identifying sarcastic examples and limiting false positive predictions among the evaluated models. GPT-OSS-20B follows with an accuracy of 0.811 and F1-score of 0.500. Kimi-K2 achieves the highest ROC-AUC (0.782), indicating the strongest ability among the evaluated models to distinguish sarcastic from non-sarcastic examples across different classification thresholds, despite its lower accuracy (0.684).

A notable failure pattern appears across several models. Mistral-Saba (Recall = 0.994), Llama-3.3-70B and Gemma-3-27B-IT (both Recall = 0.987), and DeepSeek-V3.2 (Recall = 0.975) achieve near-perfect recall while maintaining very low precision (0.192–0.301) and accuracy below the majority-class baseline. This combination indicates a strong tendency to classify examples as sarcastic: although these models identify nearly all sarcastic examples, they also incorrectly classify a large number of non-sarcastic examples as sarcastic. The resulting behavior highlights the difficulty of maintaining a balance between sarcasm recall and false-positive control in the evaluated colloquial Egyptian Arabic dataset.

D. Cross-Task Analysis

Cross-task lexical performance (ROUGE-1) across summarization and question answering is compared in Fig. 3.

The cross-task analysis yields two main findings:

- 1) *No Universally Dominant Architecture*: No single model leads across all three tasks. While Llama-3.3-70B leads in summarization lexical metrics, it drops to seventh on sarcasm detection. Gemma-3-27B-IT ranks second in question answering but falls to eighth on sarcasm detection.
- 2) *Scale Inversions*: Parameter count does not correlate directly with downstream Arabic performance. GPT-OSS-20B outperforms the larger GPT-OSS-120B on sarcasm detection (0.811 vs. 0.758 Accuracy), and the 27B Gemma-3 model outperforms 70B+ architectures on factual question answering. Pretraining data composition, tokenization, and instruction alignment appear to influence Arabic performance more than raw parameter scale [5], [13].

TABLE II

COMPARATIVE PERFORMANCE ON THE ARABIC TEXT SUMMARIZATION DATASET ACROSS LEXICAL (ROUGE-1, ROUGE-L, BLEU, METEOR) AND SEMANTIC (BERTScore-F1, SEMANTIC MATCH) METRICS. BOLD INDICATES THE BEST RESULT PER COLUMN; UNDERLINE INDICATES SECOND-BEST.

Model	ROUGE-1	ROUGE-L	BLEU	METEOR	BERTScore-F1	Semantic Match
Llama-3.3-70B	0.3587	0.2912	0.1258	0.2930	0.6680	0.618
GLM-5	<u>0.3579</u>	<u>0.2718</u>	<u>0.1205</u>	0.3080	0.6577	<u>0.908</u>
Gemma-3-27B-IT	0.3456	0.2605	0.1031	0.2908	<u>0.6665</u>	0.865
GPT-OSS-120B	0.3400	0.2490	0.1008	<u>0.3037</u>	0.6653	0.913
GPT-OSS-20B	0.3389	0.2493	0.0936	<u>0.2978</u>	0.6604	0.858
DeepSeek-V3.2	0.3333	0.2469	0.0922	0.2774	0.6626	0.860
Mistral-Saba	0.3299	0.2644	0.0975	0.2605	0.6585	0.599
Qwen3-32B	0.3181	0.2436	0.0852	0.2586	0.6544	0.823
Kimi-K2	0.3034	0.2341	0.0742	0.2362	0.6477	0.814

TABLE III

COMPARATIVE PERFORMANCE ON THE ARABICAQA DATASET ACROSS LEXICAL (ROUGE-1, ROUGE-L, BLEU, METEOR) AND SEMANTIC (BERTScore-F1, SEMANTIC MATCH) METRICS. BOLD INDICATES THE BEST RESULT PER COLUMN; UNDERLINE INDICATES SECOND-BEST.

Model	ROUGE-1	ROUGE-L	BLEU	METEOR	BERTScore-F1	Semantic Match
GLM-5	0.6095	<u>0.6046</u>	<u>0.2394</u>	0.5502	<u>0.7851</u>	<u>0.853</u>
Gemma-3-27B-IT	<u>0.6094</u>	0.6055	0.2403	<u>0.5379</u>	<u>0.7726</u>	0.840
DeepSeek-V3.2	<u>0.6069</u>	0.6029	0.2385	<u>0.5335</u>	0.7864	0.861
Llama-3.3-70B	0.5946	0.5911	0.2282	0.5212	0.7794	0.823
Kimi-K2	0.5941	0.5901	0.2270	0.5137	0.7786	0.826
Mistral-Saba	0.5871	0.5830	0.2303	0.5238	0.7795	0.825
GPT-OSS-20B	0.5776	0.5743	0.2157	0.4902	0.7613	0.840
GPT-OSS-120B	0.5771	0.5733	0.2154	0.5020	0.7655	0.842
Qwen3-32B	0.5511	0.5476	0.1981	0.4779	0.7444	0.844

TABLE IV

CLASSIFICATION PERFORMANCE ON THE AR_SARCASM DATASET ACROSS ACCURACY, MACRO-F1, PRECISION, RECALL, AND ROC-AUC. BOLD INDICATES THE BEST RESULT PER COLUMN; UNDERLINE INDICATES SECOND-BEST.

Model	Accuracy	F1	Precision	Recall	ROC-AUC
GLM-5	0.829	0.538	0.462	0.643	0.753
GPT-OSS-20B	<u>0.811</u>	0.500	<u>0.425</u>	0.607	0.728
GPT-OSS-120B	0.758	0.465	0.357	0.665	0.720
Qwen3-32B	0.749	<u>0.501</u>	0.365	0.798	0.769
Kimi-K2	0.684	0.480	0.324	0.924	0.782
DeepSeek-V3.2	0.640	0.460	0.301	0.975	<u>0.776</u>
Llama-3.3-70B	0.572	0.422	0.268	<u>0.987</u>	0.741
Gemma-3-27B-IT	0.429	0.353	0.215	<u>0.987</u>	0.656
Mistral-Saba	0.340	0.322	0.192	0.994	0.606

VI. DISCUSSION AND PRACTICAL IMPLICATIONS

A. Task-Driven Model Routing

The results show that no single evaluated model achieves the highest performance across all tasks and evaluation metrics. Model performance varies according to the task and the metric used for evaluation. In summarization, Llama-3.3-70B achieves the highest scores on ROUGE-1, ROUGE-L, BLEU, and BERTScore-F1, whereas GPT-OSS-120B achieves the highest Semantic Match score. In question answering and sarcasm detection, different models achieve the highest scores on different evaluation metrics. These results indicate that model selection can be made according to the specific task



Fig. 3. Cross-task lexical performance (ROUGE-1) comparison between Arabic Text Summarization and Question Answering across the nine evaluated models.

and evaluation objective rather than assuming that one model will perform best across all tasks.

B. Parameter Scaling and Dialectal Sensitivity

The comparison of GPT-OSS-20B and GPT-OSS-120B shows that the larger model does not consistently outperform the smaller model across the evaluated tasks and metrics. In particular, GPT-OSS-120B achieves a higher Semantic Match score than GPT-OSS-20B in summarization, while GPT-OSS-20B achieves stronger performance on the evaluated

sarcasm detection metrics. Therefore, within this evaluation, model parameter count alone does not consistently predict task performance. The results also demonstrate substantial variation in performance on the Arabic sarcasm detection task, highlighting the importance of evaluating models on the specific Arabic variety and task for which they are intended.

C. Diagnostic Value of Dual Metrics

The summarization results demonstrate that lexical overlap and semantic similarity can produce different assessments of model outputs. GPT-OSS-120B, for example, achieves the highest Semantic Match score despite ranking lower on ROUGE-1 than Llama-3.3-70B. Conversely, Llama-3.3-70B achieves the strongest scores across several lexical and embedding-based metrics while receiving a comparatively lower Semantic Match score. These results show that no single metric captures all aspects of generated-text quality. Using lexical metrics together with semantic-oriented measures such as BERTScore and LLM-as-Judge provides complementary information about the relationship between generated and reference texts [19], [20].

D. Limitations and Reproducibility Considerations

Several constraints of this study should be considered when interpreting the results:

- *API-Based Inference:* Model inference was performed through cloud-based API gateways, primarily OpenRouter and Groq. This avoids the need for local inference hardware but introduces external dependencies, including potential latency variation, rate limits, service availability, and changes to hosted model configurations.
- *Prompting Strategies:* All models were evaluated using zero-shot prompting to maintain a consistent evaluation protocol. The results therefore do not measure how model performance may change under few-shot prompting, chain-of-thought prompting, or task-specific prompt optimization. These strategies remain potential directions for future investigation [11].
- *Dialectal Scope:* The sarcasm detection evaluation uses an Egyptian Arabic dataset and therefore does not establish performance across other Arabic varieties. Results may differ for Levantine, Gulf, Maghrebi, or other regional varieties [10], [12].
- *Reference Dependence:* ROUGE, BLEU, METEOR, and BERTScore evaluate generated text in relation to reference texts. Consequently, their scores reflect similarity to the available references and may not fully capture the quality of valid responses that use substantially different wording or structure. The LLM-as-Judge evaluation provides an additional semantic perspective but also depends on the evaluation rubric and the behavior of the judge model.
- *Statistical Significance:* The rankings reported in Tables II–IV are based on aggregate metric values and should therefore be interpreted descriptively. Small differences, such as the 0.0008 ROUGE-1 difference between Llama-3.3-70B and GLM-5 in summarization, may not represent

statistically significant performance differences. Statistical significance testing, such as paired bootstrap resampling [39], is therefore planned as a complementary analysis alongside the release of per-sample benchmark results.

- *Model and API Versioning:* The benchmark results reflect the model versions and API configurations available during the evaluation period. Changes to model versions, inference settings, provider implementations, or API behavior may affect the reproducibility of the reported results. Recording model identifiers, generation parameters, and evaluation dates is therefore important for future replication.

Based on the findings of this benchmark, several practical implications emerge for Arabic NLP system development. First, model performance should be evaluated separately for each target task rather than assuming that a single model will perform consistently across different tasks. Second, model size alone should not be treated as a sufficient indicator of expected performance, as the evaluated models show different strengths across tasks and metrics. Third, generative tasks benefit from evaluating both lexical overlap and semantic similarity, since these measures can produce substantially different assessments of the same outputs. Fourth, classification models should be evaluated using class-sensitive metrics such as Macro-F1, Recall, Precision, and ROC-AUC, particularly when the class distribution is imbalanced. Finally, evaluation on data that reflects the intended Arabic variety and task is important before drawing conclusions about model suitability for a specific application.

VII. CONCLUSION AND FUTURE WORK

This paper presented an empirical benchmark of nine multilingual Large Language Models across three Arabic NLP tasks: text summarization, question answering, and sarcasm detection. Evaluating more than 22,000 instances using lexical, embedding-based, and LLM-based semantic metrics, the results show substantial variation in model performance across tasks and evaluation criteria. The comparison of different model sizes also shows that larger parameter counts do not consistently correspond to higher performance across the evaluated tasks. In summarization, lexical and semantic evaluation produced different assessments for several models, demonstrating the value of combining complementary evaluation metrics rather than relying on a single metric. The accompanying React.js benchmarking application provides a framework for evaluating candidate models on custom datasets using the same general evaluation pipeline.

Future work will extend the benchmark to additional Arabic-focused models, including Jais [15], ALLaM [24], and AceGPT [23]. The evaluation will also be expanded to cover a broader range of Arabic dialects using resources such as AraDiCE [26]. Additional experiments will investigate how few-shot prompting and other prompting strategies affect model performance across the evaluated tasks, with particular attention to their effects on dialectal language understanding and task-specific accuracy. Further work will also include statistical significance testing and more extensive per-sample

analysis to better characterize the differences observed between models.

ACKNOWLEDGMENT

The authors thank **Interact Technology Solutions** for providing computational resources and financial support for the development and evaluation of this benchmarking framework. The authors also acknowledge the developers and organizations responsible for making the evaluated models and datasets available for research and evaluation.

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